Lab Report: Comparing CNN architectures

1 Introduction

In this report we compare several models on standard benchmarks. The dataset was split into 80 percent training data and the rest for testing. The learning rate was set to 0.01 with the Adam optimizer.

2 Results

Figure 1 shows the f1 score of LSTM by dataset.

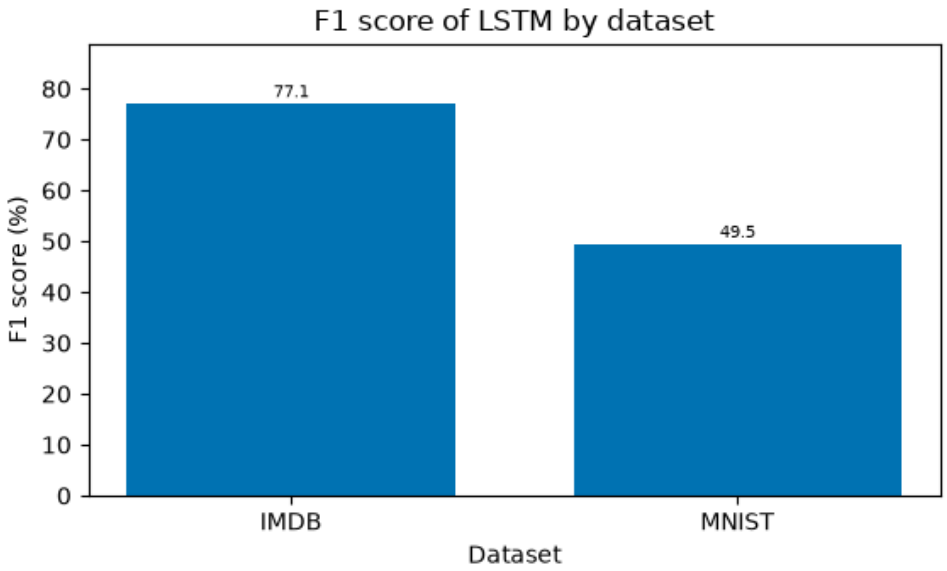


Figure 1: F1 score of LSTM by dataset for the models in this experiment.

As shown in Figure 1, LSTM achieves 86.0% F1 score on IMDB. LSTM reaches 77.1% F1 score on IMDB (Figure 1). LSTM scores 71.9 points higher on IMDB than on MNIST (Figure 1). On MNIST, LSTM obtains a F1 score of about 50%.

Figure 2 shows the validation accuracy by dataset.

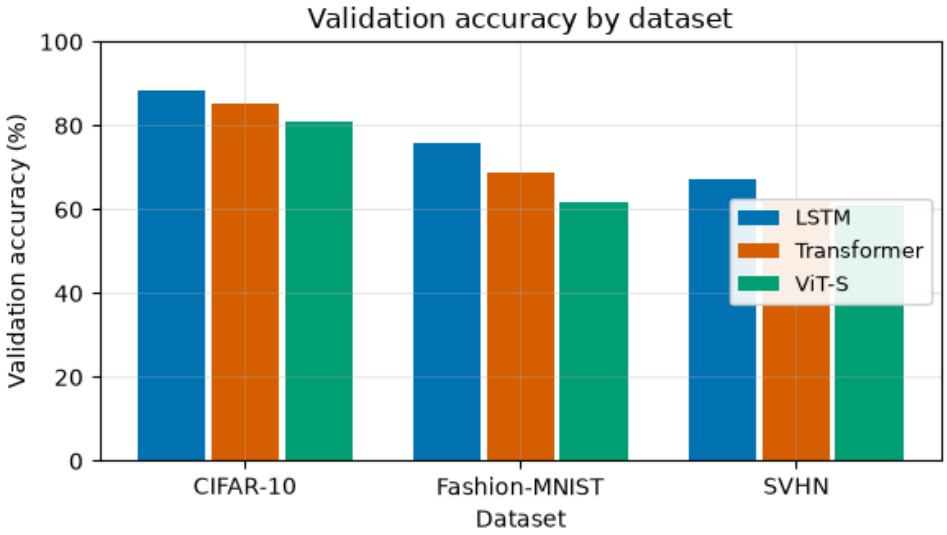


Figure 2: Validation accuracy by dataset for the models in this experiment.

On SVHN, LSTM is 10.8 points better than Transformer. As shown in Figure 2, LSTM outperforms Transformer by 5.0 percentage points on CIFAR-10.

Figure 3 shows the f1 score during training.

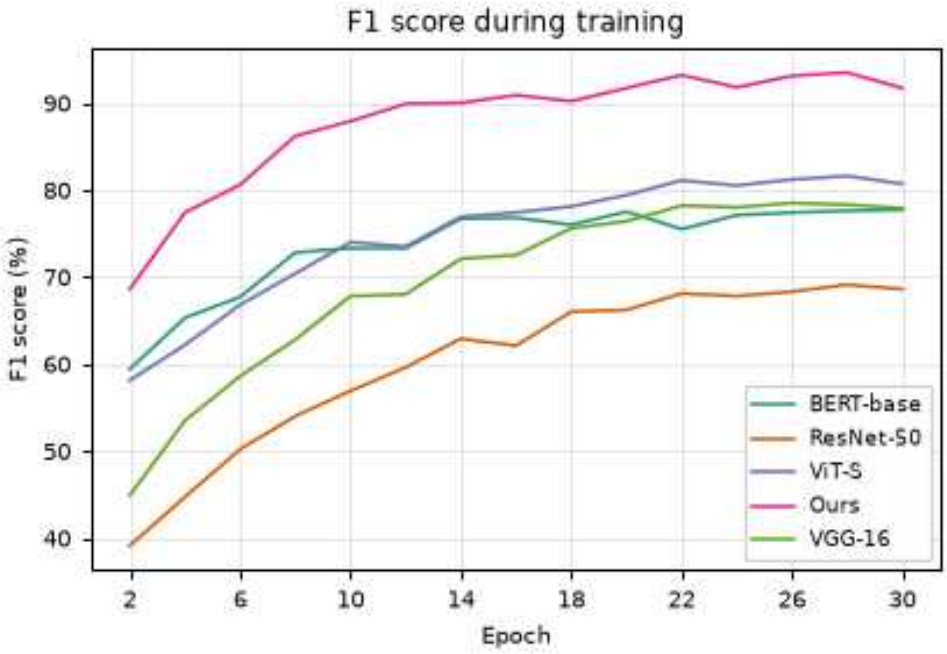


Figure 3: F1 score during training for the models in this experiment.

Ours outperforms VGG-16 by about 15 percentage points at epoch 22. From epoch 4 to epoch 14, the F1 score of BERT-base improves by a relative 35.8%.

Figure 4 shows the validation accuracy during training.

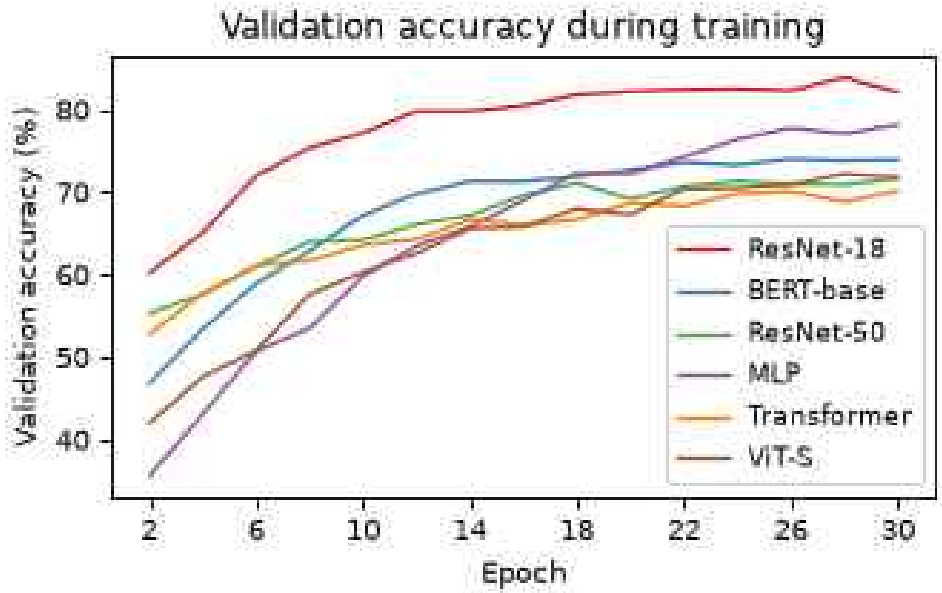


Figure 4: Validation accuracy during training for the models in this experiment.

At epoch 6, Transformer is 37.0 points better than MLP. From epoch 6 to epoch 16, the validation accuracy of MLP improves by a relative 41.3%. BERT-base reaches 69.8% validation accuracy at epoch 12.

3 Discussion

Logistic Regression reaches 64.5% accuracy on CIFAR-10. The differences between the models are discussed in the next section.